

Propulsion for Commercial Blended Wing Body Aircraft

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Abstract: The blended wing body (BWB) offers substantial fuel burn reductions relative to tube-and-wing transport aircraft, but it removes the underwing and rear-fuselage engine locations on which current propulsion integration practice rests. We review six propulsion families for commercial BWB aircraft of approximately 200 to 300 or more passengers: conventional and ultra-high bypass turbofans, open rotors, boundary layer ingestion (BLI) and distributed propulsion, hybrid-electric and turboelectric systems, and hydrogen. Published data from NASA, industry programmes and academic studies are assessed against five criteria: mission fuel burn, community noise, integration difficulty, in-flight emissions and technology readiness. Padded high-bypass turbofans above the aft centre body are the only option compatible with entry into service in the 2030s, and the airframe, rather than the engine, supplies most of the near-term efficiency gain. Centre-body shielding makes the open rotor viable, raising its cumulative noise margin to Stage 4 from approximately 13 EPNdB in a rear-fuselage installation to 30.8 EPNdB. BLI with conventional turbomachinery yields a demonstrated benefit of 3 to 5%, whereas the 70% reduction predicted for turboelectric distributed propulsion relies on superconducting machines that are not yet flight-ready. Battery storage is unsuitable as a propulsive energy source at this scale, and hydrogen, although well matched to the internal volume of the BWB, is not expected before the 2040s. We conclude by proposing a three-generation roadmap that introduces propulsion technology progressively alongside the airframe.

Index Terms: Blended wing body, boundary layer ingestion, hydrogen aircraft, open rotor, turboelectric distributed propulsion.

NOMENCLATURE

Abbreviation	Definition
BLI	Boundary layer ingestion
BPR	Bypass ratio
BWB	Blended wing body
EPNdB	Effective perceived noise level in decibels
HWB	Hybrid wing body (NASA designation for the BWB class)
LH ₂	Liquid hydrogen
OEI	One engine inoperative
SAF	Sustainable aviation fuel
TeDP	Turboelectric distributed propulsion
TRL	Technology readiness level
UHB	Ultra-high bypass

I. INTRODUCTION

A. Background

The blended wing body (BWB) integrates the fuselage and wing into a single lifting configuration. Because the centre body contributes to lift, the wetted area per passenger is reduced and the cruise lift-to-drag ratio is increased relative to a conventional tube-and-wing aircraft. Reported fuel burn benefits, however, are highly sensitive to the choice of reference aircraft. Airbus estimated a reduction of up to 20% for its MAVERIC configuration relative to a current single-aisle aircraft fitted with the same engines [1], whereas JetZero claims a reduction of up to 50% for its Z4 relative to the in-service fleet it is intended to replace, again with conventional engines [2]. The disparity arises principally from the selected baseline rather than from any difference in airframe performance.

B. Implications for Propulsion Integration

The BWB removes both locations that engines conventionally occupy on transport aircraft: beneath the wing, where adequate ground clearance is available, and on the rear fuselage. The propulsion integration problem is therefore altered in several respects. The configuration presents three principal opportunities:

- 1) a broad upper aft surface on which engines can be acoustically shielded from observers on the ground;
- 2) a thick centre-body boundary layer that can be ingested to reduce aircraft drag;
- 3) a large internal volume suited to low-density fuels such as liquid hydrogen (LH₂).

It also imposes four principal constraints:

- 1) a thrust line located well above the centre of gravity, which couples changes in thrust to the pitching moment;
- 2) the absence of a conventional vertical tail, so that the one engine inoperative (OEI) condition must be controlled by drag devices, split elevons or differential thrust;
- 3) non-uniform inflow to inlets mounted on the upper surface;
- 4) restricted maintenance access to engines mounted high on the airframe.

C. Objectives and Organisation

The objective of this paper is to establish which propulsion systems are best suited to commercial BWB aircraft and when each could plausibly enter service. Specifically, we 1) define the integration constraints common to all propulsion options on

a BWB, 2) evaluate each propulsion family against a consistent set of criteria using published data, and 3) propose a propulsion roadmap that matches technology maturity to successive aircraft generations.

Section II describes the review methodology, including its scope, sources and evaluation criteria. Section III presents the integration constraints and the findings for each propulsion family. Section IV interprets these findings through a comparative assessment, and Section V states the conclusions, the proposed roadmap and directions for future work.

II. METHODOLOGY

A. Scope

This review considers commercial BWB aircraft with capacities of approximately 200 to 300 or more passengers. The NASA designation hybrid wing body (HWB) describes the same class of configuration and is treated here as equivalent. Six propulsion families are examined: conventional and ultra-high bypass (UHB) turbofans, open rotors, boundary layer ingestion (BLI) and distributed propulsion, hybrid-electric and turboelectric systems, and hydrogen.

B. Sources

The review draws on three classes of published source: NASA technical reports and conference papers, principally those concerning hybrid wing body noise assessment and the N2B and N3-X concepts; academic studies, including AIAA and ISABE conference papers, journal articles and a doctoral thesis; and manufacturer and trade press reports on current industry programmes, including the JetZero Z4, CFM RISE, Rolls-Royce UltraFan and Airbus ZEROe. Industry sources are used principally to establish programme status and stated performance targets. All online sources were accessed on October 6, 2026.

C. Evaluation Criteria

Each propulsion family is evaluated against the five criteria defined below.

1) Mission Fuel Burn

Fuel burn benefits are reported as stated in the source. Because the sources adopt different reference aircraft and engines, the figures indicate the direction and approximate magnitude of each benefit rather than a strict ranking. Where the data permit, we separate the contribution of the propulsion system from that of the airframe.

2) Community Noise

Noise is expressed as the cumulative margin to FAA Stage 4 in effective perceived noise decibels (EPNdB), a larger margin denoting a quieter aircraft. The NASA N+2 goal of 42 EPNdB serves as a reference.

3) Integration Difficulty

Integration difficulty is assessed qualitatively against the constraints set out in Section III-A: acoustic installation, longitudinal trim, OEI yaw control, and safety and maintainability.

4) In-Flight Emissions

In-flight emissions are characterised by the energy carrier: kerosene, sustainable aviation fuel (SAF) or hydrogen.

5) Technology Readiness

Readiness is expressed as an estimated technology readiness level (TRL) together with a plausible date of entry into service on a BWB. Both are our own estimates, informed by the programme status reported in the cited sources.

III. RESULTS

A. Propulsion-Airframe Integration Constraints

On a BWB, the location of the engines is as consequential as their type. Most published commercial BWB studies locate the engines on the upper aft surface of the centre body, and this choice governs community noise, flight control and maintainability. The principal placement options identified in the literature are summarised in Table I.

TABLE I
ENGINE PLACEMENT OPTIONS FOR COMMERCIAL BWB AIRCRAFT

Placement	Example	Principal benefit	Principal penalty
Podded on pylons above aft body	JetZero Z4 [2], Boeing BWB-450	Use of existing engines; uniform inlet flow	Pylon and nacelle drag; nose-down thrust moment
Semi-buried, close to surface	Most BLI concept studies	Partial BLI; improved shielding	Inlet distortion; more demanding certification
Fully embedded with S-duct inlets	Cambridge-MIT SAX-40 [3]	Lowest noise; maximum BLI	Heavy ducts; fan blade-off hazard near cabin
Distributed fans along trailing edge	NASA N3-X [4]	Largest drag reduction	Requires electrical power transmission
Beneath the outer wing	Rarely studied	Conventional maintenance access	Limited ground clearance; no shielding

1) Acoustic Shielding

Engines mounted above the centre body are acoustically shielded from observers on the ground. NASA experiments demonstrated jet noise reductions of up to approximately 5 EPNdB when the engines were relocated two fan diameters upstream of the trailing edge [5]. Shielding effectiveness increases with frequency and decreases with angle from the jet inlet: Thomas *et al.* measured jet noise reductions of up to 8 dB at high frequencies and upstream angles, but almost none at angles beyond 130° [6]. Fan and turbomachinery noise therefore benefit more from shielding than low-frequency jet mixing noise. A system noise assessment of a BWB with turbofans representative of NASA N+2 technology predicted a cumulative margin of 34 EPNdB to Stage 4, increasing to 41.6 EPNdB when projected airframe noise reduction technology was included [7].

2) Longitudinal Trim

Because the thrust line lies well above the centre of gravity, any change in thrust produces a pitching moment that must be

trimmed by the elevons, incurring trim drag. Lowering the nacelles towards the surface reduces the moment arm but increases the proportion of boundary layer ingested by the inlet.

3) Yaw Control With One Engine Inoperative

Following an engine failure, the yawing moment is the product of the thrust of the operative engine and its lateral offset from the centre of gravity, augmented by the windmilling drag of the failed engine. Without a vertical tail, the BWB must generate the restoring moment with winglet rudders, split drag (crocodile) flaps or differential thrust. Wildschek *et al.* showed that winglet rudders alone were insufficient for the OEI case on a BWB airliner with two top-mounted turbofans, because flutter constraints limit winglet size, and that outboard crocodile flaps were also required [8]. The OEI condition is therefore a dominant sizing case for yaw control devices, which favours engine locations close to the aircraft centreline.

4) Safety and Maintainability

Installing engines above the pressurised cabin introduces rotor burst and fire zone considerations that do not arise on conventional aircraft, and engines mounted high on the airframe require access platforms for line maintenance. Conversely, an elevated intake is considerably less susceptible to ingestion of foreign objects from the runway surface.

B. Turbofans

Podded turbofans mounted above the aft centre body constitute the only propulsion option that could be certified on a commercial BWB within the present decade. The configuration is, moreover, more accommodating of large-diameter fans than the conventional tube-and-wing layout.

1) Derivative Engines

The full-scale JetZero Z4 demonstrator is to be powered by two Pratt & Whitney PW2040 turbofans, the powerplant of the Boeing 757, installed in Collins Aerospace nacelles [2]. The production aircraft requires greater thrust, and updated variants of the GE CF6 and Pratt & Whitney PW4000 are under evaluation. JetZero contends that the 45,000 ft (13.7 km) cruise altitude and the shielding provided by the centre body permit the use of updated legacy engines, provided that they satisfy current emissions standards [9]. These engines have bypass ratios (BPR) of approximately 5 to 6 (indicative values), considerably lower than those of current single-aisle engines. The efficiency gain of such an aircraft is therefore attributable almost entirely to the airframe.

2) Geared and Ultra-High Bypass Turbofans

The Rolls-Royce UltraFan technology demonstrator was first run in May 2023. It employs a geared fan architecture, has been operated at a power of 64 MW, and is claimed to deliver

a 10% efficiency improvement over the Trent XWB [10]. No launch application has been announced. On a tube-and-wing aircraft, fan diameter is constrained by landing gear length; when the engine is mounted above a BWB centre body this constraint is removed, and nacelle mass and drag become the limiting factors.

3) Installation Considerations

Three installation effects were identified. First, engines located aft of the cabin displace the centre of gravity rearwards, a shift to which the short longitudinal moment arm of the BWB is sensitive; larger and heavier fans exacerbate this effect. Second, nacelles above the body are immersed in flow accelerated by the centre body, which raises the local Mach number and interference drag, so pylon height must be traded against the degree of boundary layer ingestion. Third, a reduction in engine count simplifies the OEI case only if the engines are located close to the centreline.

C. Open Rotor

Of the propulsion families considered, the open rotor, now also termed the open fan, gains most from BWB installation, because the airframe can shield the noise that has precluded its use on airliners since the 1980s.

1) Performance

The CFM RISE open fan dispenses with the fan nacelle, allowing the fan diameter to increase without the mass and drag penalties of a duct. Its bypass ratio exceeds 60 [11], and the programme targets a fuel burn reduction of more than 20% relative to engines currently in service [12]. The fan blades rotate at less than half the speed of a narrowbody turbofan fan, and an adaptive third stream allows the effective bypass ratio to be varied in flight [13]. As of July 2026, full-size fan hardware is under test and a flight demonstration is planned within the decade [12].

2) Influence of BWB Installation on Noise

As reported in [14], an open rotor installed on the rear fuselage of a tube-and-wing aircraft achieves a cumulative margin of only about 13 EPNdB to Stage 4, which compares unfavourably with modern turbofans. Guo and Thomas assessed a BWB with three open rotors on the upper surface at approximately 24 EPNdB below Stage 4, limited by practical constraints on engine location and by main landing gear noise [14]. For a hybrid wing body freighter with three open rotors, Thomas *et al.* obtained 26.0 EPNdB with the rotors 1.0 diameter upstream of the trailing edge and 30.8 EPNdB at 1.5 diameters, and estimated that 38.0 EPNdB could be reached with further noise reduction technology [15]. These results are compared with turbofan BWB configurations in Fig. 1.

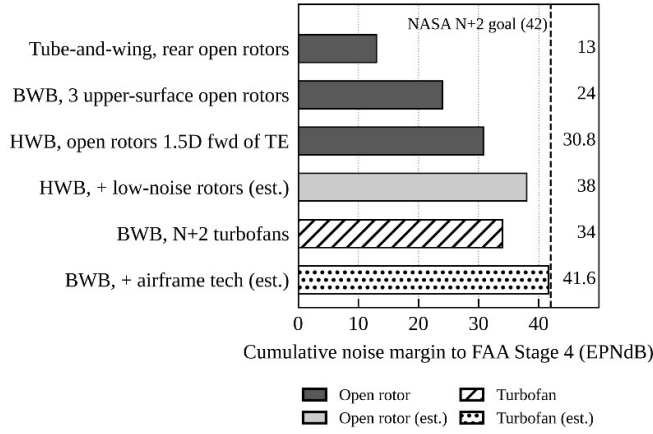


Fig. 1. Cumulative noise margin to FAA Stage 4 for open rotor and turbofan configurations [7], [14], [15]. The dashed line denotes the NASA N+2 goal of 42 EPNdB; paler bars denote estimates that include projected technology.

Relocating the open rotor from a rear-fuselage installation to the upper surface of a hybrid wing body increases the cumulative margin by approximately 18 EPNdB. Turbofan-powered BWBs nevertheless remain quieter, and only the configuration that includes projected airframe noise technology approaches the NASA N+2 goal.

3) Unresolved Issues

Four issues remain unresolved. First, in the absence of a containment casing a released blade is unconstrained, so locating rotors above or behind a pressurised cabin makes blade-release protection a central certification issue. Second, large-diameter rotors require tall pylons on a BWB, which add structural mass and increase the nose-down thrust moment. Third, cabin noise and vibration from unducted blades in close proximity to the passenger deck have not been characterised for this configuration. Fourth, the installed efficiency of the open rotor remains contested, because the elimination of nacelle drag is partially offset by pylon and interference losses.

D. Boundary Layer Ingestion and Distributed Propulsion

BLI is the propulsion concept most naturally suited to the BWB. With conventional turbomachinery, however, the demonstrated benefit is a few per cent, considerably smaller than the reductions sometimes quoted for the concept.

1) Principle

A propulsor that ingests low-momentum air from the airframe surface requires less shaft power to produce a given thrust, and it partially re-energises the aircraft wake. The broad, flat upper surface of the centre body develops a thick boundary layer over a span equal to many fan diameters, which is why distributed propulsors along the trailing edge are particularly well suited to the BWB.

2) Reported Benefits

Table II summarises the fuel burn benefits reported for BWB and HWB concepts incorporating BLI. Because the baselines differ, the figures are not directly comparable.

TABLE II
REPORTED FUEL BURN BENEFITS OF BLI AND DISTRIBUTED PROPULSION CONCEPTS

Concept	Propulsion system	Reported fuel burn reduction	Baseline
NASA N3-X [4]	Two turboshaft-driven superconducting generators; 15 motor-driven BLI propulsors	70 to 72%	Boeing 777-200LR
NASA HWB, podded UHB geared turbofans [4]	Two pylon-mounted engines	52%	Boeing 777-200LR
Cambridge-MIT SAX-40 [3]	Three cores driving nine embedded geared fans	23%	Current civil aircraft
NASA BLI2DTF [16]	Embedded BLI inlet with distortion-tolerant fan	3 to 5%	Large HWB with podded engines
Cranfield BWB study [17]	Distributed fans with BLI	5.3%	Turbofan BWB

The two NASA configurations form the most informative pair. At an identical airframe and technology level, the podded configuration achieved a 52% reduction, so the turboelectric distributed propulsion (TeDP) system accounts for the additional 18 to 20 percentage points by eliminating nacelle and pylon drag, maximising BLI and reducing wake drag [4]. This result, however, assumes superconducting generators, motors and transmission lines that are far from flight readiness. The 3 to 5% reduction for which the BLI2DTF wind tunnel test demonstrated the required component performance [16] is a more representative guide for conventional machinery.

3) Inlet Distortion

A BLI fan encounters non-uniform inflow during every revolution. Such distortion presents aeromechanical challenges, notably flutter stability and forced-response dynamic stresses [18]. Kim and Liou found that the offset BLI inlets of the NASA N2B configuration were susceptible to cowl flow separation and shock formation, and achieved a 12.5%

reduction in a circumferential distortion indicator through adjoint-based inlet shape optimisation [19]. The Cranfield study likewise found the distributed propulsion system to be particularly sensitive to inlet duct losses [17].

E. Hybrid-Electric and Turboelectric Propulsion

For a BWB of 200 to 300 seats, batteries cannot store a meaningful fraction of the propulsive energy required. Electrical power transmission nevertheless remains valuable, because it enables distributed propulsion and mitigates the OEI problem.

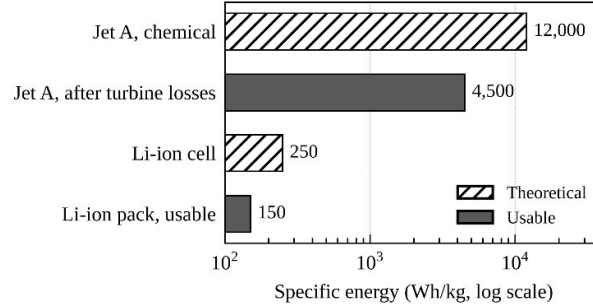


Fig. 2. Specific energy of Jet A and lithium-ion storage. Useful values account for turbofan losses and for battery pack integration and drivetrain losses, respectively [20].

2) Candidate Architectures

Three architectures were identified, in order of increasing electrification.

Mild hybrid: A motor-generator on a turbofan shaft provides assistance during take-off and climb, permitting a smaller core sized for cruise. CFM is ground testing hybrid-electric systems within the RISE programme [12]. The benefit is modest, but so is the technical risk.

Partial turboelectric: Conventional engines produce most of the thrust and also drive generators that power a small number of BLI propulsors at the trailing edge, capturing part of the BLI benefit.

Full turboelectric: Gas turbines drive generators only, and all thrust is produced by distributed electric propulsors. On the N3-X, two wingtip turboshaft engines drive superconducting generators that supply 15 motor-driven propulsors [4], each turbogenerator producing more than 30 MW [21]. Superconducting or cryogenically cooled machines are required to achieve an acceptable system mass.

3) Engine-Out Benefit for Tailless Aircraft

If each turbine supplies propulsors on both sides of the aircraft, the loss of one turbine reduces thrust symmetrically and produces a negligible yawing moment [22]. For a tailless BWB, in which the OEI condition sizes the yaw control devices (Section III-A3), this constitutes an advantage with no equivalent on tube-and-wing aircraft.

F. Hydrogen

The BWB is the most suitable airframe for LH₂, because it provides the internal volume that tube-and-wing aircraft lack.

1) The Energy Storage Gap

Jet A has a specific energy of approximately 12,000 Wh/kg. After the thermal efficiency of the turbofan core and component losses are taken into account, slightly more than one third of this energy propels the aircraft, giving a useful specific energy of approximately 4,500 Wh/kg [20]. Lithium-ion cells reach approximately 250 Wh/kg; integration into flight-capable packs reduces this to below 200 Wh/kg, and an assumed drivetrain efficiency of 75% yields a useful specific energy of approximately 150 Wh/kg [20]. As Fig. 2 shows, the resulting ratio of approximately 30 precludes battery-powered cruise for a long-range airliner. Batteries, moreover, do not lose mass as they discharge, so the landing mass remains high.

Hydrogen propulsion itself, however, is not now expected before the 2040s.

1) Storage Volume

LH₂ has approximately 2.8 times the specific energy of kerosene but requires approximately four times the volume for the same stored energy (indicative property values). Tanks must be insulated cylindrical or spherical vessels that maintain the fuel at approximately 20 K, so they cannot be integrated into wing boxes. A tube-and-wing aircraft must therefore sacrifice cabin length to accommodate them, whereas a BWB can locate them within the deep centre body adjacent to or aft of the cabin.

2) Industry Status

In 2020 Airbus presented three ZEROe hydrogen concepts, including a BWB of up to 200 seats powered by two hybrid-hydrogen turbofans, and identified its wide interior as offering multiple options for hydrogen storage and distribution [23]. In February 2025 Airbus deferred the programme by five to ten years and reduced its budget by approximately one quarter, citing slower than expected technology maturation [24].

3) Combustion and Fuel Cell Routes

The two principal means of converting hydrogen to thrust are compared in Table III.

TABLE III
COMPARISON OF HYDROGEN PROPULSION ROUTES FOR A BWB

Route	Propulsor	Advantages	Disadvantages
Hydrogen combustion	Modified turbofan or open rotor	Established hardware; high power density	Continued NO _x emission; increased water vapour, hence persistent contrail formation
Hydrogen fuel cell	Electric propulsors	No in-flight CO ₂ or NO _x ; compatible with distributed BLI propulsors	High stack mass; large heat rejection; limited power at present

TABLE IV
COMPARATIVE ASSESSMENT OF PROPULSION OPTIONS FOR COMMERCIAL BWB AIRCRAFT

Option	Fuel burn benefit (engine or propulsion only)	Noise on a BWB	Integration difficulty	In-flight CO ₂	Readiness (TRL, est.)	Plausible BWB service entry
Legacy turbofan, BPR 5 to 6	None; gain provided by airframe	Good with shielding	Low	Kerosene or SAF	9	Early 2030s
Geared / UHB turbofan	Approx. 10% relative to Trent XWB	Very good	Low to medium	Kerosene or SAF	6 to 7	Mid 2030s
Open rotor	Over 20% relative to engines in service	Acceptable only with shielding	Medium to high (blade release over cabin)	Kerosene or SAF	5	Late 2030s
Partial BLI, close-mounted fans	3 to 5%	Very good	Medium (fan distortion)	Kerosene or SAF	4 to 5	Late 2030s
Mild hybrid-electric	Small; permits a smaller core	Unchanged	Low to medium	Kerosene or SAF	5 to 6	2030s
Turboelectric distributed BLI	Additional 18 to 20 percentage points (N3-X)	Excellent	High (MW-scale electrics, cooling)	Kerosene, SAF or H ₂	2 to 3	2045+
Hydrogen combustion	Not comparable on a fuel mass basis	Similar to turbofan	High (cryogenic tanks)	Zero	4	2040s
Hydrogen fuel cell	Not comparable on a fuel mass basis	Excellent	Very high	Zero	3	2045+

The fuel burn values in Table IV are taken from the sources cited in Section III and are referenced to different baselines. They therefore indicate the direction and approximate magnitude of each benefit rather than a strict ranking.

B. Near-Term Options

A geared or UHB turbofan is the most credible propulsion choice for a BWB entering service in the 2030s. Because the airframe provides most of the fuel burn reduction (Section III-B1), a mature engine of moderate bypass ratio represents a reasonable first step, consistent with the approach adopted by JetZero [9]. Mild hybridisation is a low-risk near-term addition to any turbofan or open rotor BWB.

C. Medium-Term Options

The open rotor is the most promising medium-term option, with entry into service plausible in the late 2030s. The results in Fig. 1 indicate that centre-body shielding largely neutralises its principal disadvantage, although turbofan-powered BWBs remain quieter. Its viability therefore depends less on noise than on whether blade-release protection over a pressurised cabin can be satisfactorily resolved. Partial BLI with closely coupled turbofans is a credible enhancement for a second-generation BWB, provided that distortion-tolerant fans reach maturity.

The N3-X concept burns LH₂ in its turbogenerators, in part because the cryogenic fuel can serve as a heat sink for the superconducting motors and transmission lines [21].

IV. DISCUSSION

A. Comparative Assessment

No single propulsion system is superior against every criterion. The principal trade-off lies between near-term certainty and long-term efficiency, and the BWB shifts this trade-off by relaxing the noise constraint that governs engine selection on tube-and-wing aircraft. Table IV consolidates the findings of Section III against the criteria of Section II-C.

D. Long-Term Options

Fully distributed BLI requires lightweight and efficient power transmission, either mechanical, as on the SAX-40, or electrical, as on the N3-X, and is not expected before the 2040s. Full turboelectric distributed propulsion is the long-term ideal, but it depends on megawatt-class, flight-weight electrical machines and thermal management systems that do not yet exist. Hydrogen combustion in a turbofan or open rotor is the most plausible zero-CO₂ propulsion route for a large BWB. It is unlikely to enter service before the 2040s, and its introduction is constrained more by fuel production, airport infrastructure and tank mass than by the engine itself.

On a BWB these long-term technologies are mutually reinforcing: LH₂ provides a heat sink for superconducting power transmission, which in turn enables distributed BLI propulsors that exploit the thick centre-body boundary layer and provide symmetric thrust following a turbine failure.

E. Limitations

Three limitations qualify these conclusions. First, the fuel burn figures are referenced to differing baselines and cannot be ranked strictly. Second, several figures, notably those for the JetZero Z4 and CFM RISE, are manufacturer claims or programme targets rather than independently verified results.

Third, the TRL values and entry-into-service dates are our own estimates and will change as programmes such as RISE and the Z4 engine selection progress.

V. CONCLUSION

This review has shown that the most suitable propulsion system for a commercial BWB depends on its entry-into-service date, and that propulsion technology should be introduced progressively alongside the airframe rather than simultaneously with it. We therefore propose the following three-generation roadmap.

- 1) *First generation (2030s): two podded high-bypass turbofans above the aft centre body.* A certified or derivative engine should be installed close to the centreline and approximately two fan diameters upstream of the trailing edge to maximise acoustic shielding. This approach excludes engine development risk from a programme that already carries airframe, cabin and flight control risk. A geared or UHB turbofan in the appropriate thrust class is the natural subsequent upgrade.
- 2) *Second generation (late 2030s to 2040s): open rotor or closely coupled BLI turbofans, with mild hybridisation.* The open rotor offers the largest single efficiency gain, and the BWB largely neutralises its noise penalty. Blade-release protection over the cabin is the first problem to be resolved. Partial immersion of the nacelles in the boundary layer offers a further gain of a few per cent if distortion-tolerant fans reach maturity.
- 3) *Third generation (2045 onwards): turboelectric distributed BLI propulsion, preferably fuelled by LH₂.* This architecture realises the full potential of the BWB: it eliminates pylon and nacelle drag, re-energises the wake, provides symmetric thrust following a turbine failure and exploits the internal volume available for cryogenic tanks. Its realisation depends on flight-weight, megawatt-class electrical machines and on an established hydrogen supply chain.

Future work should address four issues as the field develops: 1) certification of engines installed above a pressurised cabin, particularly with respect to uncontained rotor failure; 2) the trim drag penalty associated with a high thrust line, which offsets part of the cruise benefit; 3) inlet distortion and its effect on fan durability in any BLI installation; and 4) the selection of the production engine for the JetZero Z4, expected within approximately one year of mid-2026, which may determine the first-generation choice in practice [9].

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